

Experimentally Testing Bell's Inequality and Single Photon Tomography

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The tension between local realism and quantum mechanics, first formalized by the EPR paradox and later by Bell's inequality, remains a cornerstone of quantum information science. While violating Bell's inequality confirms nonlocality, it provides only a partial snapshot of the underlying quantum state. A complete description requires reconstructing the system's density matrix to quantify its purity and entanglement. In this paper, we report the generation and characterization of polarization entangled photon pairs via spontaneous parametric down-conversion (SPDC). Using the quTools entanglement demonstrator and performing polarization correlation measurements, we demonstrate a clear violation of the Clauser-Horne-Shimony-Holt (CHSH) inequality, thereby confirming the presence of nonlocal correlations. To further probe the system's coherence, we perform quantum-state tomography on the constituent photons. This dual approach allows us not only to witness non-classicality through Bell-parameter evaluation but also to quantitatively map the state's superposition via density matrix reconstruction.

I. INTRODUCTION

Physicists Albert Einstein, Boris Podolsky, and Nathan Rosen proposed a thought experiment in 1935[1], in which a source emits a pair of entangled particles traveling in opposite directions. Moreover, EPR argued that these two particles must satisfy the criteria of local realism, in which they are predicted to possess some definite physical quantity prior to measurement, and are fully independent once no longer interacting. Since quantum mechanics appeared to allow correlations incompatible with these assumptions, EPR declared the theory incomplete.

29 years later, John Bell formulated a theorem applying local realism to two entangled particles, traditionally photons or electrons, in which he assumes a local hidden variable λ , correlating the states of both particles and constrained by locality[2]. Bell showed that any local hidden variable model must satisfy this set of inequalities bounding the strength of correlations between separated particles. Quantum mechanics, as it turns out, predicts violations of these inequalities for certain entangled states. As such, experimental observations of a Bell inequality violation must rule out local realistic descriptions, and prove the presence of nonclassicality[3].

Throughout this paper, we will primarily consider polarization entangled photon pairs. Our polarization measurements follow the Clauser-Horne-Shimony-Holt (CHSH) formulation of Bell's inequality[4], which we derive below.

II. THEORY

A. Bell's Inequality

The distribution of these pairs, emitted with respect to the hidden variable λ , is represented by the probability

distribution $\rho(\lambda)$, in which:

$$\rho(\lambda) \geq 0 \quad \text{and} \quad \int \rho(\lambda) d\lambda = 1 \quad (1)$$

This given pair of entangled particles, A and B, is characterized with respect to λ as $A(\lambda, a) = \pm 1$ at detector 1 (in orientation a) and $B(\lambda, b) = \pm 1$ at detector 2 (in orientation b). This correlation of these two orientations of a and b is presented by the function:

$$E(\mathbf{a}, \mathbf{b}) = \int d\lambda \rho(\lambda) A(\lambda, \mathbf{a}) B(\lambda, \mathbf{b}) \quad (2)$$

From these correlations across different orientations, we can form S, the total measure of correlation between a and b, represented by correlation functions or CHSH inequalities, such as $E(\mathbf{a}, \mathbf{b})$. Considering all contributing correlations to the overall total measurement, we write:

$$S(a, a', b, b') = E(a, b) - E(a, b') + E(a', b) + E(a', b') \quad (3)$$

In this calculation, as S is described by the four polarization correlation coefficients: planes a and b in addition to their perpendicular components a' and b'. In this total correlation, S is bounded by the local hidden variable such that:

$$S(\lambda, a, a', b, b') \leq \pm 2 \quad (4)$$

However, predictions from Quantum Mechanics for entangled particles contradict this statement as:

$$S_{QM} = 2\sqrt{2} \quad (5)$$

From this violation of S, the local hidden variable is thus obsolete. To obtain this violation of Bell's inequality, the correct reference plane is essential, as different orientations yield different violations of Bell's inequality. We can obtain the maximum violation utilizing:

$$S_{QM}(\mathbf{a}, \mathbf{b}, \mathbf{a}', \mathbf{b}') = \cos(\mathbf{a}, \mathbf{b}) - \cos(\mathbf{a}, \mathbf{b}') + \cos(\mathbf{a}', \mathbf{b}) + \cos(\mathbf{a}', \mathbf{b}') \quad (6)$$

To simplify and solve the equation, we utilize the relations: $(a, b') = (a, b) + (b, a') + (a', b')$, $(a, b) = (b, a') = (a', b') = \theta$, and $\sin(\theta) = \sin(3\theta)$. Utilizing these, we arrive at the maximum and minimum violations of S_{QM} in respect to their angle of measurement:

$$S_{QM} = 2\sqrt{2} \quad \text{for} \quad \theta = \pm \frac{\pi}{8} \quad (7)$$

$$S_{QM} = -2\sqrt{2} \quad \text{for} \quad \theta = \pm \frac{3\pi}{8} \quad (8)$$

Considering a case of entangled photons, these would be the following orientations of our polarizers to analyze orientations of horizontally and vertically polarized photons to obtain the largest violations of Bell's inequality[5].

B. Tomography

A violation of Bell's inequality serves as a demonstration of non-classicality, confirming that the observed correlations cannot be described by a local hidden-variable theory. However, this violation is a scalar metric; it does not uniquely identify the specific quantum state responsible for the correlations. Distinct states, ranging from pure entangled states to certain mixed states, can yield identical Bell parameters. Even more so, the experimental design may be inherently flawed, leading to a state different from the intended one. To fully characterize the polarization mode and quantify its coherence, it is therefore necessary to reconstruct the state directly through quantum state tomography.

The state reconstruction method we will employ is linear inversion, following the protocol set by quTools[6]. For reasons that will be outlined in the Results section, this discussion will pertain only to single qubit polarization, and as a result, single photon tomography.

The complete information of the photonic system is contained within the density operator ρ . For a single polarization qubit, ρ acts on the Hilbert space spanned by the horizontal and vertical basis states $|H\rangle$ and $|V\rangle$. As ρ is a 2×2 Hermitian matrix with unit trace, it can be uniquely decomposed in the basis of the Pauli spin operators σ_i :

$$\rho = \frac{1}{2} \sum_{i=0}^3 T_i \sigma_i = \frac{1}{2} (\mathbb{I} + \vec{T} \cdot \vec{\sigma}) \quad (9)$$

where $T_0 = \text{Tr}(\rho) = 1$ enforces the unit trace condition. The real coefficients $T_i = \text{Tr}(\rho \sigma_i)$ constitute the correlation tensor elements $\vec{T} = (T_0, T_x, T_y, T_z)$. This is analogous to the Stokes vector, meaning just as the

Stokes coefficients map the polarization projections to a geometric point on (or inside) the Poincaré sphere, we are mapping the quantum state to the Bloch sphere.

The Pauli operators correspond physically to measurements in three conjugate polarization bases: rectilinear (H/V), diagonal (P/M), and circular (R/L) (again, notation is chosen to conform with quTools). We determine the correlation parameters by recording the normalized intensity differences in these bases. For a basis of measured coincidence counts c , the correlation parameters are given by:

$$T_x = \frac{c_p - c_m}{c_p + c_m}, \quad T_y = \frac{c_r - c_l}{c_r + c_l}, \quad T_z = \frac{c_h - c_v}{c_h + c_v} \quad (10)$$

These correlation mappings are determined by their typical orientation on the Bloch sphere, where the Z axis is our rectilinear polarizations, X is diagonal, and Y circular.

This reconstruction allows for the direct evaluation of the state properties not accessible via the Bell parameter alone, including the degree of polarization and the state purity. Thus, while the Bell inequality establishes the non-classical nature of the correlations, tomography reveals the explicit form of the quantum state.

III. METHODS

A. Experiment Setup

We generated polarization entangled photon pairs via type-I spontaneous parametric down-conversion (SPDC) using a quTools Entanglement Demonstrator. The source utilizes a pair of β -Barium Borate (BBO) nonlinear crystals pumped by a continuous wave diode laser at $\lambda_p = 405\text{nm}$. The optical assembly includes a half-wave plate (HWP) for state preparation, sandwiched between pre- and post-compensation yttrium orthovanadate (YVO_4) crystals to mitigate temporal walk-off[7].

To produce the maximally entangled $|\Phi^+\rangle$ state, the pump polarization was set to 22.5° via the HWP, creating an equal superposition of horizontal and vertical polarization:

$$\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \quad (11)$$

The two BBO crystals are oriented with orthogonal optical axes. In the first crystal, $|V\rangle$ pump photons undergo down-conversion into horizontally polarized pairs ($|V\rangle \rightarrow |HH\rangle$), while in the second crystal, $|H\rangle$ pump photons produce vertically polarized pairs ($|H\rangle \rightarrow |VV\rangle$). Because these processes occur within a coherent volume, the pathways interfere to produce the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|HH\rangle + e^{i\phi}|VV\rangle) \quad (12)$$

where ϕ is a relative phase adjusted via the compensation crystals.

Degenerate SPDC converts the 405 nm pump photons into signal and idler pairs centered at 810 nm, constrained by energy conservation ($\hbar\omega_p = \hbar\omega_s + \hbar\omega_i$). We coupled the photons into collection optics at the intersection of the two SPDC emission cones. Precision alignment at this intersection is critical; any transverse shift favors one crystal's emission over the other. As will be discussed later, this proved to be a key challenge in the experiment.

Detection was performed using two silicon avalanche photodiodes. With the laser operated at a drive current of 37.3mA, we observed single-photon rates exceeding 10kHz and coincidence rates above 1kHz within a 40ns resolution window.

B. Bell's Inequality

To measure Bell's inequality, we use collection optics and linear polarizers, followed by an avalanche photodiode, which records the final measurement. Nonetheless, to take the following measurements, the optics setup needs to be correctly oriented to align with the center of the BBO crystal in order to obtain equal amounts of horizontal and vertical polarization.

Beginning the experiment, we take an initial measurement with no adjustment to the gauge system. If coincident counts are not ideal, we adjust the position of the avalanche photodiodes by twisting the knobs controlling their horizontal and vertical axes individually to maximize the correlation counts. The maximized correlations are taken with respect to the polarizations of 0° (all vertical), 90° (all horizontal), and 45° (mix of both states). If there are counts at least around 1300 in all frames (however, counts above 2000 are preferred), then we proceed to adjust both avalanche photodiodes simultaneously, in which they are both shifted in the same direction, to change the point of incidence within the BBO crystal with respect to the fixed position of the mirrors. If this proves unsuccessful as well, we proceed to alter the mirrors. Before making any changes, we label their original orientations as a point of reference, then from this, we make a shift either to the left or the right, and then proceed to redo the two previous steps of changing the orientation of the avalanche photodiodes to maximize the counts again. These steps are then repeated until the coincidences are maximized.

After reaching the required coincidence count in all frames, we can take measurements using the equations listed in the theory of Bell's State, as well as some new ones. On one polarizer, we take measurements from 0° , 90° , and 45° , and -45° to obtain a mix of the photon polarizations in all frames. For the other polarizer, we take measurements at 22.5° , 67.5° , 112.5° , and 157.5° . Using these measurements, we obtain the angles of maximized violations of Bell's Inequality, in addition to their perpendicular components. After taking all of the following measurements and intermixing angles from each

polarizer, we use the following equation to compute the correlation function by counting coincidences within the following angles of measurement.

$$E(\alpha, \beta) = \frac{C(\alpha, \beta) - C(\alpha, \beta_\perp) - C(\alpha_\perp, \beta) + C(\alpha_\perp, \beta_\perp)}{C(\alpha, \beta) + C(\alpha, \beta_\perp) + C(\alpha_\perp, \beta) + C(\alpha_\perp, \beta_\perp)} \quad (13)$$

Then, using these same counts, we are able to compute the Gaussian error with:

$$E(a, b) = 2 \frac{[C(a, b) + C(a_\perp, b_\perp)][C(a, b_\perp) + C(a_\perp, b)]}{(C(a, b) + C(a, b_\perp) + C(a_\perp, b) + C(a_\perp, b_\perp))^2} \times \sqrt{\frac{1}{C(a, b) + C(a, b_\perp)} + \frac{1}{C(a, b_\perp) + C(a_\perp, b)}} \quad (14)$$

After obtaining all of the following correlation functions and their errors, we are able to calculate the S value with the equation:

$$S(a, a', b, b') = |E(a, b) + E(a', b) - E(a, b') + E(a', b')| \quad (15)$$

This measured S value acts as the final check of the effectiveness of our alignment. If all steps are successful, then the S value will be above 2, providing a direct violation of Bell's inequality [8]

C. Tomography

To characterize the source, we performed state tomography using the collection optics, a quarter-wave plate (QWP), and a linear polarizer followed by an avalanche photodiode. While the Bell tests prove two-photon correlations, single qubit tomography reconstructs the polarization state density matrix of a single arm through projections into the horizontal/vertical (H/V), diagonal/antidiagonal (P/M), and right/left circular (R/L) bases.

A systematic phase error was introduced by the only available QWP, which was designed for $\lambda_0 = 694\text{nm}$ rather than the $\lambda = 810\text{nm}$ signal wavelength. We modeled the effective phase retardance $\delta(\lambda)$ as scaling inversely with wavelength[9]:

$$\delta(\lambda) = \frac{\lambda_0 \pi}{\lambda 2} \quad (16)$$

For our operating wavelength, the retardance was $\delta \approx 1.35$ rad, deviating from the ideal $\frac{\pi}{2} \approx 1.57$ rad ($\approx \Delta 12^\circ$). This discrepancy results in imperfect conversion between linear and circular states. To mitigate this, we experimentally calibrated the QWP by identifying the angular orientations corresponding to intensity extrema through the polarizer, defining these as our effective 0° and 45° reference axes. The QWP was mechanically stabilized within the optical train to ensure alignment consistency across all measurements.

TABLE I. Single-photon tomography measurement settings. Angles are measured in degrees relative to the fast axis of each optic.

Count Basis	QWP Angle (°)	Polarizer Angle (°)
C_H	0	0
C_V	0	90
C_P	0	45
C_M	0	135
C_R	45	90
C_L	45	0

The specific angular configurations for these projections are summarized in Table I. Single photon rates are taken at each configuration.

The two-photon tomography procedure extends this methodology by replicating the projection optics in the second detection arm. To fully characterize the two-qubit density matrix, joint coincidence measurements (rather than single rates) are recorded across all combinations of the six single-photon bases. This results in an expanded measurement set of $6 \times 6 = 36$ unique configurations, providing the necessary correlation data to reconstruct the state of the entangled pair.

IV. RESULTS

A. Bell's Inequality

We performed Bell's inequality, measuring the correlation functions between various angles corresponding to maximum violations of Bell's inequality and various combinations of horizontally and vertically polarized light. From these values, we obtained the following coincidences for the two sessions of measurements:

TABLE II. Coincidence Counts for Bell's Inequality Experiment

Session	Angle	Linear Polarizer Angle			
		α	α'	$\alpha+90$	$\alpha'+90$
4/3	$\beta = 22.5$	1350	1030	120	400
	$\beta' = 67.5$	250	610	410	75
	$\beta+90 = 112.5$	280	70	380	580
	$\beta'+90 = 157.5$	1300	475	85	850
4/10	$\beta = 22.5$	1100	1000	240	430
	$\beta' = 67.5$	200	950	1180	450
	$\beta+90 = 112.5$	250	400	1100	990
	$\beta'+90 = 157.5$	1130	430	200	950

From these coincidences, we calculated the following correlation functions and their corresponding Gaussian errors for the angles set on our linear polarizers.

From the calculations in Tables III and IV, there is a clear difference in S values, as we first measured $S = 2.3$, while in the second session $S = 2.1$. Looking more closely

TABLE III. Correlation Coefficients, Errors, and Bell Parameter – April 3

	$E(\alpha, \beta)$	$E(\alpha', \beta)$	$E(\alpha, \beta')$	$E(\alpha', \beta')$
Value	0.6244	0.5481	-0.6724	0.4527
Error	0.0169	0.0184	0.0256	0.0199
$S = 2.2976$	$\sigma_S = 0.0409$			

TABLE IV. Correlation Coefficients, Errors, and Bell Parameter – April 10

	$E(\alpha, \beta)$	$E(\alpha', \beta)$	$E(\alpha, \beta')$	$E(\alpha', \beta')$
Value	0.6357	0.4113	-0.7048	0.3669
Error	0.0149	0.0172	0.0334	0.0176
$S = 2.1187$	$\sigma_S = 0.0440$			

at Table II, we observe that the lower S value corresponds to lower coincidence counts measured on April 10. This correlation between the coincidence count and the S value highlights the influence of a well-aligned system. Specifically, if the mirrors are misaligned with respect to the BBO crystals, there is an imbalance between the $|HH\rangle$ and $|VV\rangle$ components, which further results in a drop in the coincidence count. Regardless, the calculated S values verify the violation of Bell's inequalities, further disproving local realism in entangled photons.

B. Tomography

We performed single photon tomography to contrast a pure polarization state with the reduced state of an entangled pair. By measuring the projections in the H/V , P/M , and R/L bases, we reconstructed the correlation parameters (Bloch/Stoke vector components T_i) and the corresponding density matrices ρ .

TABLE V. Single-photon tomography measurements performed with and without the pump half-wave plate (HWP). Uncertainties correspond to Gaussian counting errors.

Basis	Without HWP	With HWP
C_H	103250 ± 300	5100 ± 300
C_V	8300 ± 100	49000 ± 300
C_P	77500 ± 300	52000 ± 200
C_M	3900 ± 200	48000 ± 200
C_R	7300 ± 200	42000 ± 200
C_L	98600 ± 200	57000 ± 200
T_x	0.33 ± 0.005	0.04 ± 0.003
T_y	-0.86 ± 0.004	-0.15 ± 0.003
T_z	0.85 ± 0.002	0.02 ± 0.010

In the first configuration (without the pump HWP), the photon occupied a largely pure elliptical state. Assuming gaussian noise, our measurements yielded Bloch components $T_x = 0.33 \pm 0.005$, $T_y = -0.86 \pm 0.004$, and

$T_z = 0.85 \pm 0.002$, resulting in the density matrix:

$$\rho_{pure} = \begin{pmatrix} 0.93 & 0.17 - 0.43i \\ 0.17 + 0.43i & 0.07 \end{pmatrix} \quad (17)$$

The large magnitude of the correlation tensor ($|\vec{T}| \approx 1$) and the significant off diagonal terms indicate high phase coherence and a well-defined polarization state. In this case, elliptical, about the horizontal and left circular polarizations.

In contrast, tomography performed with the HWP set for the Bell state production ($|\Phi^+\rangle$) revealed a nearly maximally mixed state. The reconstructed correlation components vanished near the origin ($T_x = 0.04, T_y = -0.15, T_z = 0.02$), yielding:

$$\rho_{mixed} = \begin{pmatrix} 0.50 & 0.02 - 0.08i \\ 0.02 + 0.08i & 0.50 \end{pmatrix} \quad (18)$$

This transition is quantified by the state purity, $\gamma = Tr(\rho^2)$, which dropped to $\gamma = 0.514$, approaching the theoretic minimum of 0.5 for a maximally mixed qubit. Furthermore, the state exhibited a fidelity of $F = 0.993$ with the ideal maximally mixed state ($I/2$). Let's now determine if this makes sense. While the global two photon state $|\Psi_{12}\rangle$ is pure, tracing over one subsystem ($\rho_1 = Tr_2[|\Psi_{12}\rangle\langle\Psi_{12}|]$) yields a mixed reduced density matrix[10]. The lack of a definite single photon polarization, as evidenced by the suppressed off-diagonals, serves as indirect evidence of the underlying non-locality.

Full characterization of these correlations requires two photon tomography to reconstruct the 4×4 joint density matrix. While we attempted this measurement, the fidelity was limited by the cumulative effect of the $\lambda = 694$ nm QWP retardance error and mechanical instabilities in the improvised mounts. Furthermore, continuous small drifts in the SPDC collection alignment between the 36 necessary measurement configurations introduced state imbalance, which prohibited a stable reconstruction of the joint density matrix. Consequently, while the single photon results confirm the expected local behavior of the source, direct measurement of the two photon concur-

rence remains a goal for future iterations with stabilized, wavelength matched optics.

V. CONCLUSION

In this work, we verified the violation of Bell's inequalities using Qu-tools entanglement setup's polarization entangled two qubit source, and characterized the resulting states through single photon tomography. Our results support the rejection of local realism in quantum systems, and demonstrate that entangled photons cannot be explained by hidden variable theories.

We found the experimental fidelity to be highly sensitive to the alignment of the collection optics relative to the center of the BBO crystal. Poor alignments result in low coincident counts and imbalance in $|HH\rangle$ and $|VV\rangle$ components, creating bias directly affecting measured S -parameters. As such, precise alignments of the collection optics to the center of the SPDC emission cones remains the most critical factor in achieving high-fidelity entanglement, as measured by S -values.

In single-qubit tomography, we compared the pure state of a non-entangled beam to the maximally mixed state produced under Bell state conditions. We observed a significant reduction in state purity ($\gamma = 0.514$). This transition from a well-defined polarization to a maximally mixed polarization matrix provides strong indirect evidence of non-locality in the source, as the individual photon subsystems lack definite properties prior to measurement.

Continuing this study, future refinements will focus on improving the mechanical stability of the tomography optics and alignment of the collection optics. Further, utilizing wavelength matched waveplates would remove phase retardance errors. This entails a more detailed and thorough walk of the beam, centering our point of incidence directly at the BBO crystals' intersection. With optimized alignment and enhanced stability, the transition from single photon to full two photon tomography will allow for the reconstruction of the joint density matrix, providing a direct measurement of the system's entanglement.

VI. REFERENCES

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